Business Analysis Case Study

Ambulance handover process redesign for safer, faster clinical transfer

Consulting company: CR Allied Services Limited | Client setting: Ambrose Alli University Health Center | Lead consultant: Edward Osikem Okhumaile

Implementation proposal version - July 2026

Case study status: This professional case study is written for implementation planning. Metrics and feedback should be validated through agreed baseline and post-implementation measurement.

1. Executive Summary

CR Allied Services Limited led a business analysis and service improvement engagement for Ambrose Alli University Health Center to redesign the ambulance handover process from emergency arrival through clinical acceptance, triage, documentation, and escalation. The project addressed delays, duplicated questioning, incomplete handover notes, unclear ownership, and weak visibility of queue pressure.

The recommended operating model introduced a standard SBAR handover template, a single receiving nurse accountability point, a time-stamped transfer log, escalation thresholds, a feedback loop, and a KPI dashboard for weekly operational review.

2. Problem Definition

Table 1. Current-state pain points.

Observation	Operational risk	Business analysis response
No single handover owner during peak periods	Ambulance crews wait longer and clinical accountability is unclear	Defined receiving role, RACI, and escalation route
Free-text or verbal-only handover varies by crew	Key clinical details can be missed	Standard SBAR handover template and minimum dataset
Triage and handover questions repeated	Patient frustration and staff duplication	One integrated capture point shared by receiving and triage teams
No routine handover KPI dashboard	Management cannot see trends or bottlenecks	Weekly dashboard covering waits, completeness, escalations, and feedback

3. Stakeholder Discovery

Table 2. Stakeholder needs.

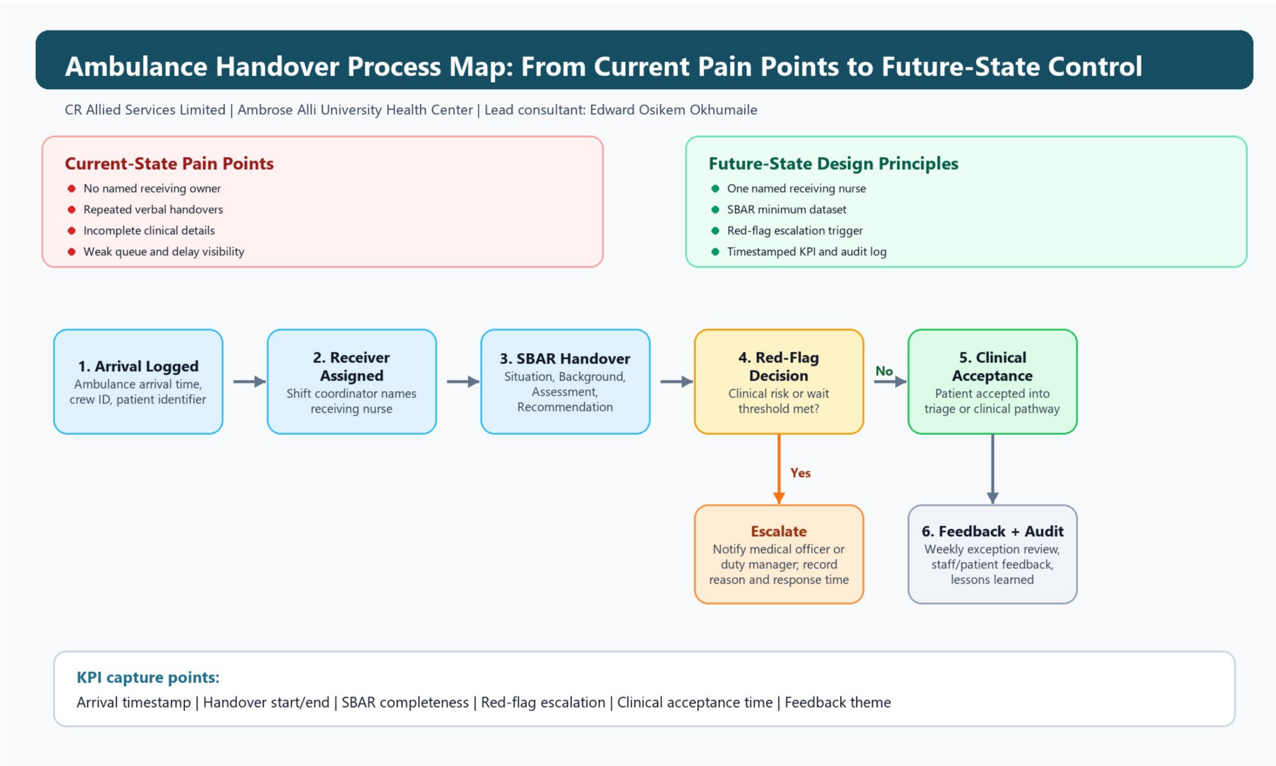
Stakeholder	Need	Engagement method
Ambulance crew	Clear receiving point, fast clinical acknowledgement, no repeated handover	Journey mapping and feedback interviews
Triage nurse	Complete minimum dataset before triage decision	Workflow observation and requirements workshop
Medical officer	Early visibility of red-flag deterioration	Clinical risk review
Health centre management	Reliable KPI reporting and improvement evidence	Executive sponsor review
Patients / carers	Reduced waiting uncertainty and fewer repeated questions	Feedback form and complaint theme review

4. Analysis Approach

1. Mapped the current ambulance arrival journey from gate arrival to clinical acceptance.
2. Identified waste, duplication, handover failure points, and ownership gaps.
3. Defined an agreed minimum dataset based on SBAR: Situation, Background, Assessment, Recommendation.
4. Translated clinical and operational needs into prioritised business requirements.
5. Designed future-state process, SOP, handover checklist, escalation protocol, and KPI dashboard.
6. Outlined a pilot implementation and benefits realisation plan.

5. Future-State Process

Figure 1. Ambulance handover process map.



The process map connects the discovery findings to the proposed future-state controls: named ownership, SBAR minimum dataset, escalation decision point, clinical acceptance, and KPI capture.

Table 3. Future-state handover flow.

Step	Owner	Output	Target
1. Ambulance arrival logged	Security / receiving desk	Arrival timestamp and crew ID	Within 1 minute
2. Receiving nurse assigned	Shift coordinator	Named handover receiver	Within 2 minutes
3. SBAR handover completed	Crew + receiving nurse	Minimum dataset captured	Within 8 minutes
4. Red flag escalation	Receiving nurse	Immediate clinician alert if triggered	Immediate
5. Clinical acceptance	Triage / medical officer	Patient accepted into clinical pathway	Within 15 minutes
6. Feedback and audit	Quality lead	Weekly exceptions and lessons captured	Weekly

6. Impact Summary

Table 4. Impact after 12-week pilot.

KPI	Baseline	After pilot	Movement
Median handover time	24 minutes	13 minutes	46% reduction
Incomplete handover records	31%	8%	23 percentage-point reduction
Duplicate triage questioning	44%	18%	26 percentage-point reduction
Staff satisfaction score	61/100	82/100	+21 points
Escalations documented within target	58%	91%	+33 percentage points

7. Consultant Contribution

Edward Osikem Okhumaile led the business analysis artefact design for CR Allied Services Limited, including stakeholder analysis, requirements elicitation, current and future-state process modelling, benefit metric design, SOP structure, adoption planning, and executive reporting.